(do not allow more than 2 sig. fig)

3 (a)	$v^2 = u^2 + 2as$ OR use of triangle etc	C1	
	$4.0^2 = 2 \times 9.8 \times s$ OR $s = ½ \times 4.0 \times 0.4$		
	$s = 0.82$ m OR 0.80 m	A1	[2]
(b)	$\Delta p = m(v - u)$ OR $p = mv$	C1	
	speeds are 4.2 m s^{-1} and 3.6 m s^{-1}	C1	
	$\Delta p = 0.045\,(4.2 + 3.6)$ (2/4 only if speeds not added)	C1	
	$= 0.35$ N s	A1	[4]
	(1 mark only if only one speed used)		
(c)	any time between 0.14 s and 0.17 s	C1	
	force $= \Delta p / \Delta t = 0.35 / 0.14$ (allow e.c.f.)		
	$= 2.5$ N	A1	[2]